Rank Sum Test - Lab Activity - Example Solution

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Checking for Shift Model vs. General Procedures

How do you tell if the shift model is reasonable? Make pictures - either univariate or overlaid for the two variables. It's easier to tell with more observations. You can also use common sense and logic. For example, scores tend to be skewed left, while salaries are skewed right. You could look for a shift in salary, and that would be reasonable (provided you thought the spread was constant). There are more general procedures that don't require shift models. So, briefly, we should consider which procedures to use when.

Consider the following sets of observations in terms of all possible pairs, and see which you think would lead you to do the shift test and which would be general procedures. Example code is provided for the first pair.

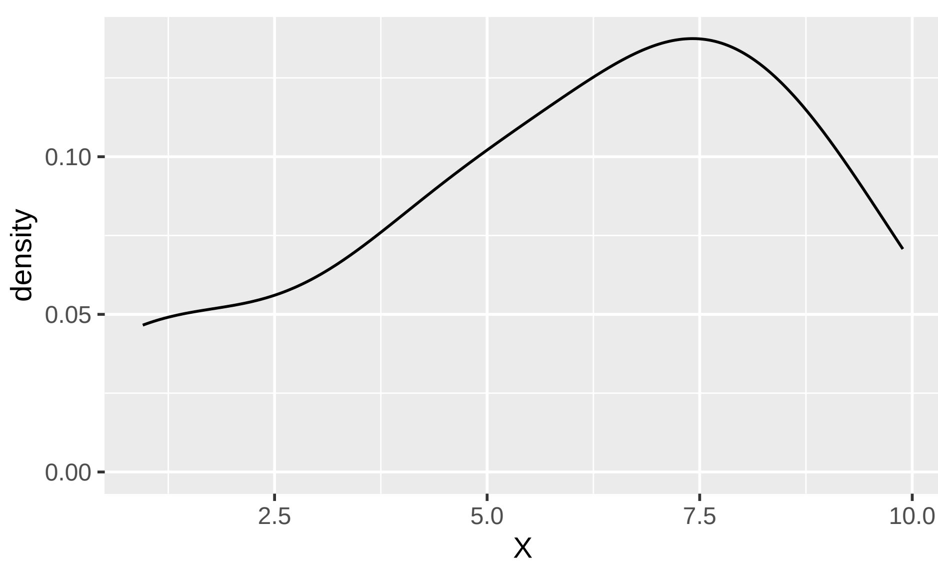
```
X <- c(4.26, 9.35, 7.96, 8.35, 9.89, 3.09, 4.79, 7.70, 7.99,
       0.99, 6.34, 0.95, 6.47, 4.99, 6.79)
Y <- c(10.72, 12.40, 7.13, 9.10, 6.78, 7.45, 5.54, 10.81, 13.95, 10.93,
       NA, NA, NA, NA, NA)
W <- c(4, 5, 6, 5, 5, 11, 7, 10, 8, 7, 8, 4, 11, 9, 12)
U <- c(1.50, 0.14, 3.04, 3.06, 8.58, 0.67, 0.19, 3.56, 6.61, 4.01,
       NA, NA, NA, NA, NA)

tmpdata <- data.frame(cbind(X, Y, W, U))
```

Would testing X vs. Y be more appropriate using a shift model or a more general procedure?

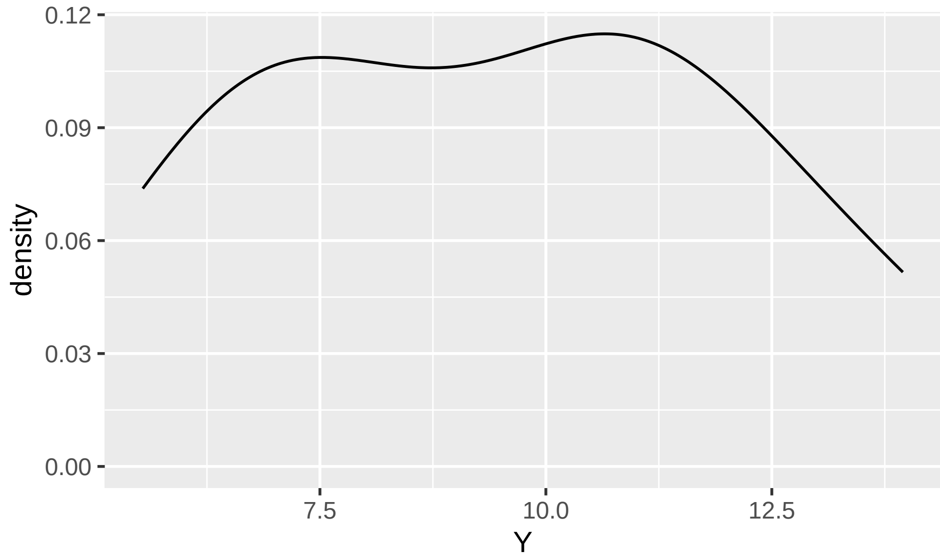
SOLUTION

```
# example code, may not need all of these plots
ggplot(data = tmpdata, mapping = aes(x = X)) + geom_density()
```



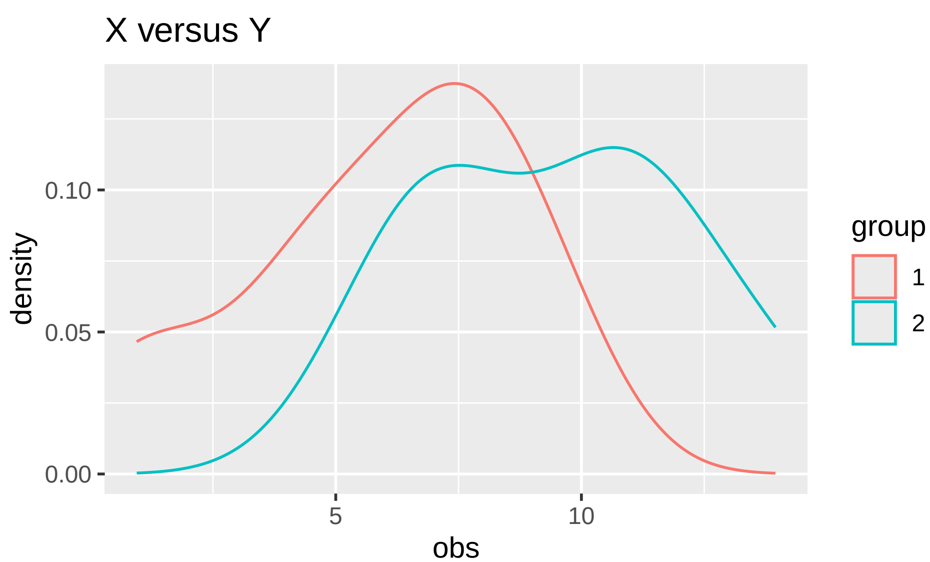
```r
ggplot(data = tmpdata, mapping = aes(x = Y)) + geom_density()
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```r
tmpdata2 <- data.frame(obs = c(X, Y),
                       group = as.factor(c(rep(1,length(X)), rep(2,length(Y)))))
ggplot(data = tmpdata2, mapping = aes(x = obs)) +
  geom_density(aes(color = group)) + labs(title = "X versus Y")
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_density()`).
```

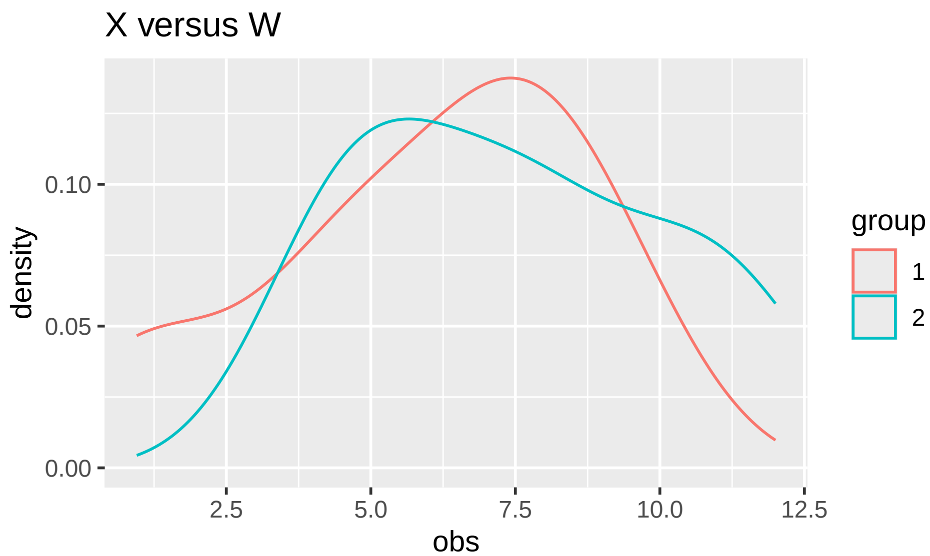


This looks like it should be a general procedure. The distributions of X and Y have opposite skews, and the Y distribution seems to have 2 peaks rather than 1 peak (X has just 1). A shift model does not appear to be appropriate.

How about X vs. W or X vs. U?

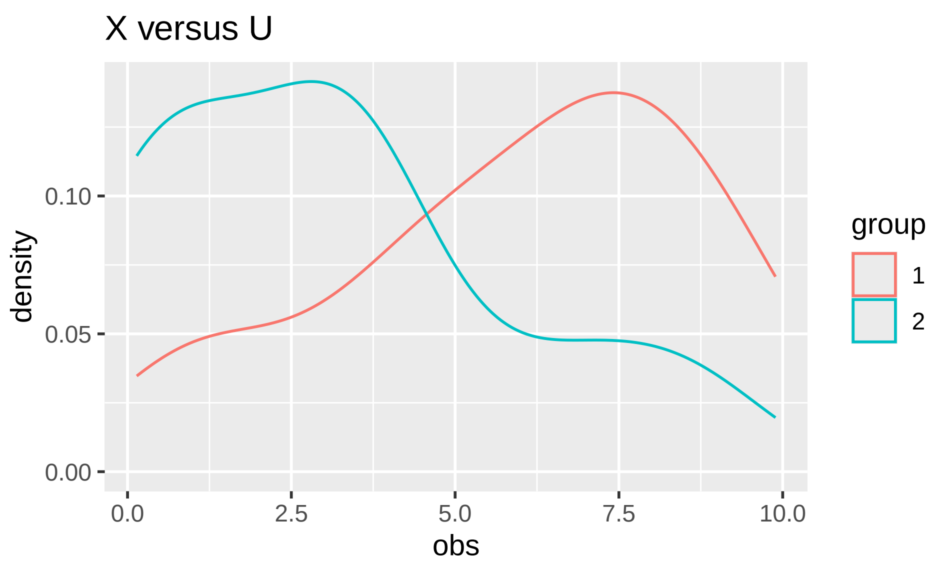
SOLUTION

```r
# X vs W
tmpdata3 <- data.frame(obs = c(X, W),
                       group = as.factor(c(rep(1,length(X)), rep(2,length(W)))))
ggplot(data = tmpdata3, mapping = aes(x = obs)) +
  geom_density(aes(color = group)) + labs(title = "X versus W")
```




```r
# X vs U
tmpdata4 <- data.frame(obs = c(X, U),
                       group = as.factor(c(rep(1,length(X)), rep(2,length(U)))))
ggplot(data = tmpdata4, mapping = aes(x = obs)) +
  geom_density(aes(color = group)) + labs(title = "X versus U")
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_density()`).
```



X vs. W seem to have opposite skew, but both with 1 peak. A shift model is more reasonable here, if we assume the skew issues are due to outliers. We'd still probably use a general procedure.

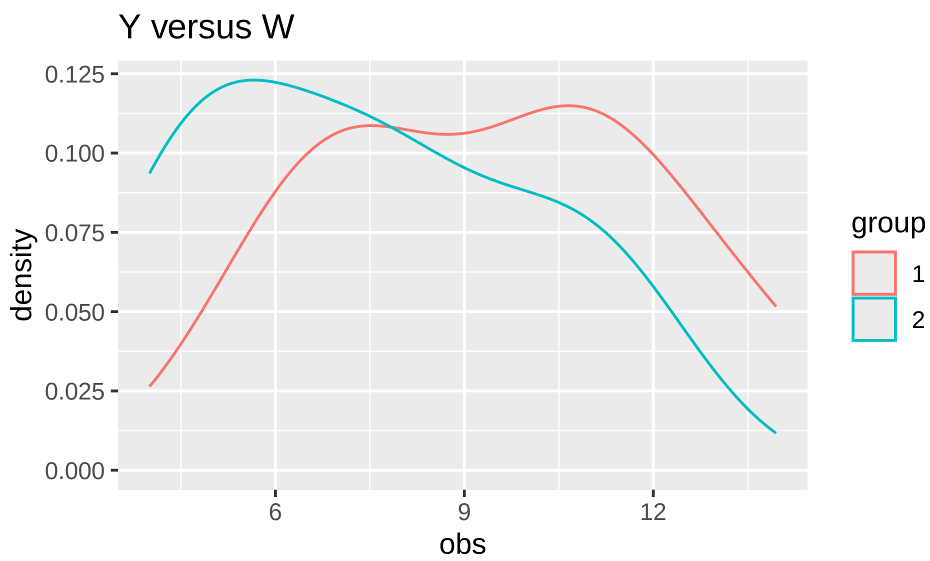
X vs. U also seem to have opposite skew, but again, both have one peak. Again, if we assume the skew issues are due to outliers, a shift model may be reasonable here.

What about Y vs. W or Y vs. U?

SOLUTION

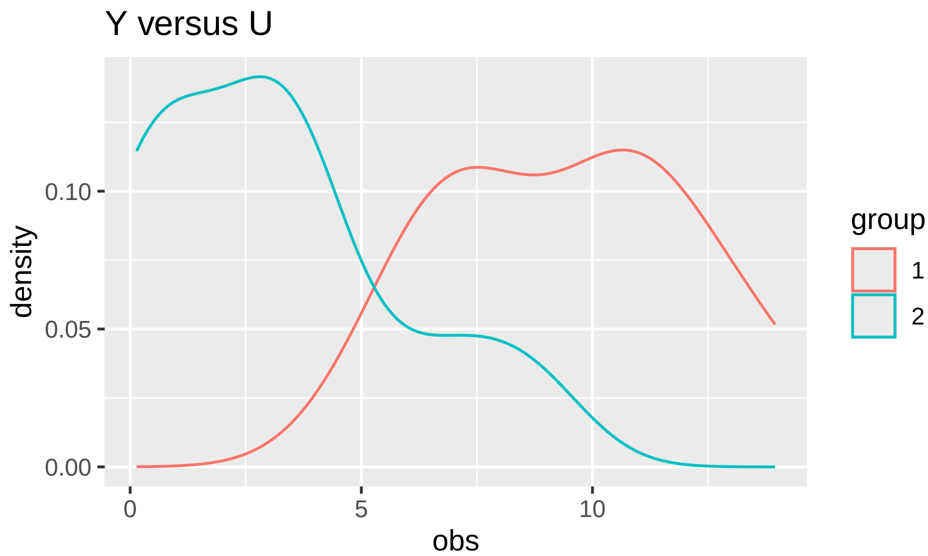
```r
# Y vs W
tmpdata5 <- data.frame(obs = c(Y, W),
                       group = as.factor(c(rep(1,length(Y)), rep(2,length(W)))))
ggplot(data = tmpdata5, mapping = aes(x = obs)) +
  geom_density(aes(color = group)) + labs(title = "Y versus W")
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```r
# Y vs U
tmpdata6 <- data.frame(obs = c(Y, U),
                       group = as.factor(c(rep(1,length(Y)), rep(2,length(U)))))
ggplot(data = tmpdata6, mapping = aes(x = obs)) +
  geom_density(aes(color = group)) + labs(title = "Y versus U")
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Y vs. W might be reasonable for a shift model, even though both have 2 peaks. Their spreads look similar. Y might be shifted above W a little bit and due to chance variation, the peaks don't quite line up in terms of their densities.

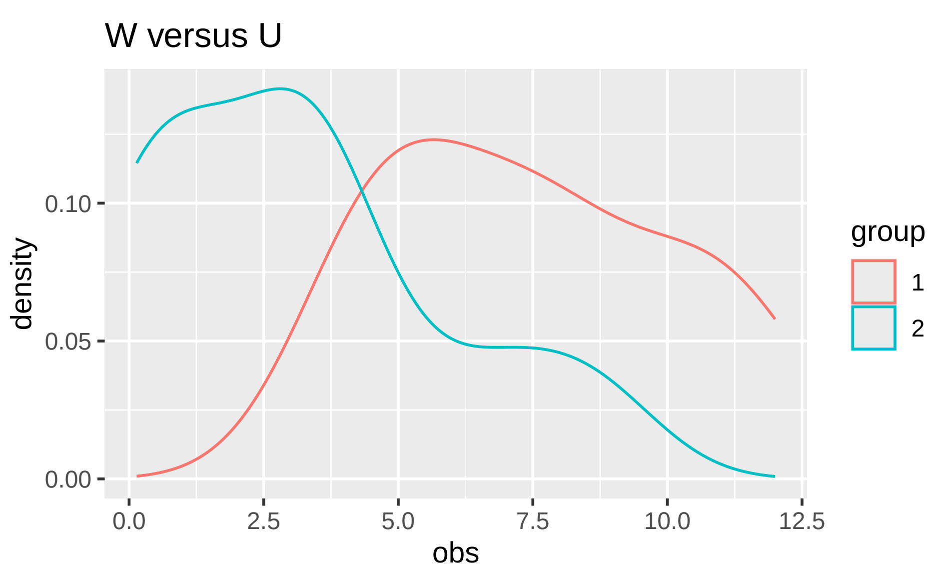
Y vs. U seems less reasonable for a shift model, so we'd want a general procedure. Y clearly has 2 peaks while U has a main one, and maybe a second small one due to outliers.

Finally, what about W vs. U?

SOLUTION

```
tmpdata7 <- data.frame(obs = c(W, U),
                       group = as.factor(c(rep(1,length(W)), rep(2,length(U)))))
ggplot(data = tmpdata7, mapping = aes(x = obs)) +
  geom_density(aes(color = group)) + labs(title = "W versus U")
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Here, both distributions show the same general pattern, but the decline is faster for U. A shift model seems somewhat reasonable here.

Note that these are subjective. The idea would be if you saw major differences in distribution shape, you'd use a general procedure. If you see shapes and spreads that are more similar, it's up to you to decide how similar is similar enough for a shift model to be reasonable. For all these generated values, the SD (spread) is between 2.65-2.78, so they have fairly similar spreads.

Example Analyses for You to Try

```
data(Day1Survey)
names(Day1Survey)
```

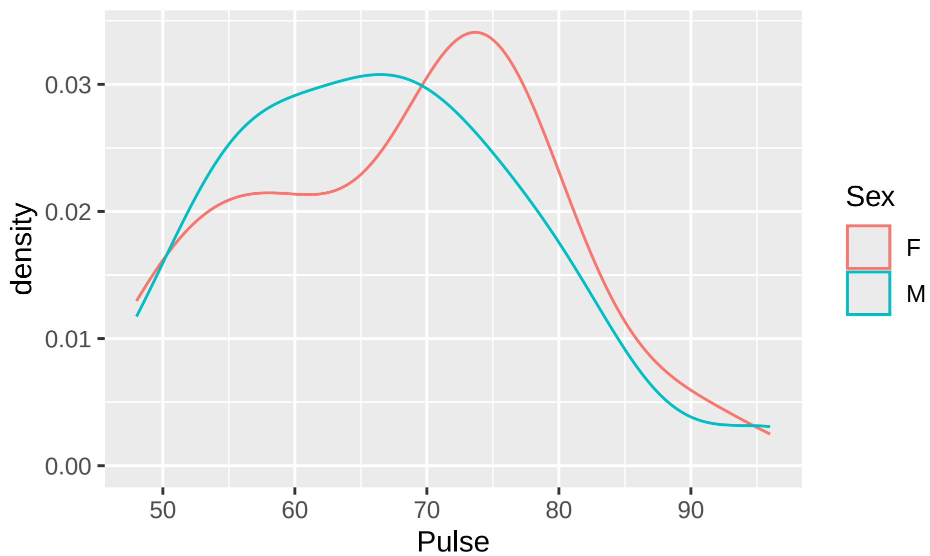
```
##  [1] "Section"     "Class"       "Sex"         "Distance"    "Height"
##  [6] "Handedness"  "Coins"       "WhiteString" "BlackString" "Reading"
## [11] "TV"          "Pulse"       "Texting"
```

Data collected on the first day of a statistics course contains many variables about the students in the course.

Does the mean/center of resting pulse differ by sex? Use appropriate procedures - one nonparametric and one parametric. Do the procedures agree?

SOLUTION

```
# code for Exploratory data analysis (EDA) provided
ggplot(data = Day1Survey, mapping = aes(x = Pulse)) +
  geom_density(aes(color = Sex))
```




```
favstats(~ Pulse | Sex, data = Day1Survey)
```

```
##   Sex min Q1 median Q3 max     mean      sd  n missing
## 1   F  51 60     72 75  90 67.82353 11.3811 17       0
## 2   M  48 57     66 72  96 66.65385 11.2710 26       0
```

```
wilcox.test(Pulse ~ Sex, data = Day1Survey)
```

```
##
##  Wilcoxon rank sum test with continuity correction
```

```
##
## data:  Pulse by Sex
## W = 238, p-value = 0.6807
## alternative hypothesis: true location shift is not equal to 0
```

```r
t.test(Pulse ~ Sex, data = Day1Survey)
```

```
##
##  Welch Two Sample t-test
##
## data:  Pulse by Sex
## t = 0.33077, df = 34.12, p-value = 0.7428
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  -6.015993  8.355360
## sample estimates:
## mean in group F mean in group M
##        67.82353        66.65385
```

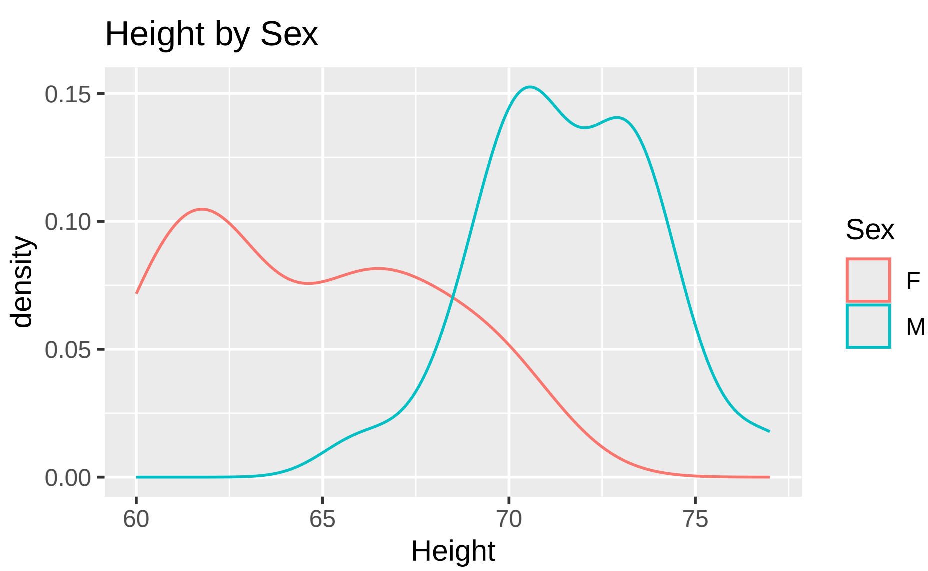
The female and male distributions of pulse have similar SDs but slightly different shapes. So, we might suggest a general procedure test, not one that uses a shift model. If you aren't as worried about the shape issue, you could use the shift model.

Both the Wilcoxon Rank Sum Test and parametric t-test have very large p-values. We do not have significant evidence that mean resting pulse differs by sex.

Does the mean/center of height differ by sex? Use appropriate procedures - one nonparametric and one parametric. Do the procedures agree?

SOLUTION

```r
# code for Exploratory data analysis (EDA) provided
ggplot(data = Day1Survey, mapping = aes(x = Height)) +
  geom_density(aes(color = Sex)) + labs(title = "Height by Sex")
```




```r
favstats(~ Height | Sex, data = Day1Survey)
```

```
##   Sex min Q1 median Q3 max     mean       sd  n missing
## 1   F  60 62     65 67  70 64.64706 3.371594 17       0
```

```
## 2   M  66 70     71 73  77 71.57692 2.386178 26       0
```

```r
wilcox.test(Height ~ Sex, data = Day1Survey)
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  Height by Sex
## W = 20, p-value = 5.469e-07
## alternative hypothesis: true location shift is not equal to 0
```

```r
t.test(Height ~ Sex, data = Day1Survey)
```

```
##
##  Welch Two Sample t-test
##
## data:  Height by Sex
## t = -7.3552, df = 26.385, p-value = 7.523e-08
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  -8.865143 -4.994586
## sample estimates:
## mean in group F mean in group M
##        64.64706        71.57692
```

The spreads look too different to probably use a shift procedure, but we try anyway to practice. Both the tests have very small p-values, and indicate that there is a significant difference in mean/center height by sex. (Technically, the one for the Wilcoxon rank sum says there is a shift in the two distributions.)

Do the shift models appear appropriate for the nonparametric procedures above? How about the conditions for the parametric methods you applied?

SOLUTION

The first test on Pulse has similar spreads, so it depends on how pronounced we think the second peak in the female distribution is. For the second test (on Height), both distributions have two peaks, but the spreads are pretty different. If the spreads were more similar, a shift model would be reasonable there.

For the parametric test conditions, these distributions are non-normal (many have two peaks), and the sample sizes are 17 and 26, respectively, which are somewhat small to assume a t-test is appropriate. So, we may have cause for concern with conditions for both test procedures here.

Second Example Analysis for You to Try

```r
data(Salaries)
# ?Salaries #remove the hashtag, i.e., the # symbol, to learn about the data
names(Salaries)
```

```
## [1] "rank"          "discipline"    "yrs.since.phd" "yrs.service"
## [5] "sex"           "salary"
```

We want to see if there is any evidence that male professors were earning more than the female professors (in the description, this is listed as the concern). Assess this, using at least two nonparametric (hint, you have two from 4.1-4.3 so far!) and one parametric procedure. Note that this analysis assumes a shift model is appropriate, and you can comment on the validity of that assumption. Here is a plot to help you get started.

```r
ggplot(data = Salaries, mapping = aes(x = salary)) +
  geom_density(aes(color = sex)) + labs(title = "Salary by Sex")
```

SOLUTION

```r
favstats(salary ~ sex, data = Salaries)
```

```
##      sex   min    Q1 median       Q3    max     mean       sd   n missing
## 1 Female 62884 77250 103750 117002.5 161101 101002.4 25952.13  39       0
## 2   Male 57800 92000 108043 134863.8 231545 115090.4 30436.93 358       0
```

The shift model may not be reasonable because the spreads are very different and the female distribution has a second peak more pronounced than the male distribution. But we can see what the tests would give us!

```r
wilcox.test(salary ~ sex, data = Salaries, alternative = "less")
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  salary by sex
## W = 5182.5, p-value = 0.004118
## alternative hypothesis: true location shift is less than 0
```

```r
vdwtest<-with(Salaries, waerden.test(salary, sex)) # be sure agricolae is loaded
vdwtest$statistics
```

```
##      Chisq Df     p.chisq
##   9.387252  1 0.002184995
```

```r
t.test(salary ~ sex, data = Salaries, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  salary by sex
## t = -3.1615, df = 50.122, p-value = 0.001332
## alternative hypothesis: true difference in means between group Female and group Male is less than 0
## 95 percent confidence interval:
##       -Inf -6620.263
```

```
## sample estimates:
## mean in group Female   mean in group Male
##             101002.4             115090.4
```

All three tests (both nonparametric and the parametric procedure) have very small p-values, (though Van der Waerden's is two-sided) and indicate that there is significant evidence that the mean/center female salary is less than the mean/center male salary.

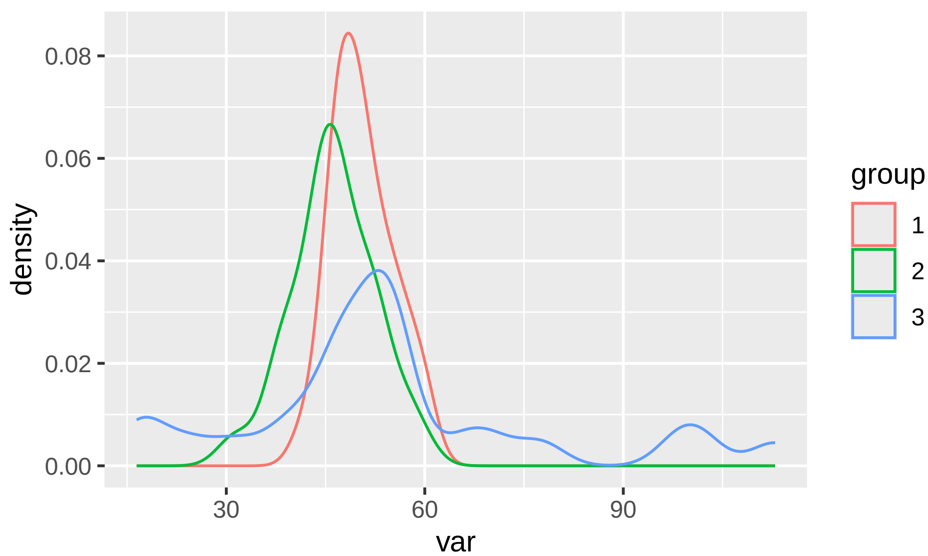
Comparing Shift Models and Test Procedures

We will generate some random data in order to investigate differences between the related test procedures for shift models and the t-test. We will also include some data that is not appropriate for a shift model to illustrate that other methods are needed.

```r
set.seed(225)
X <- rnorm(25, 50, 5)
Y <- rnorm(25, 45, 5) # shift model - centers differ, dist and spread same
Z <- rgamma(25, 3.2, 0.05) # non-shift model, center is above X's theoretically
```

Plots of the generated data

```r
var <- c(X, Y, Z)
group <- as.factor(c(rep(1:3, each = 25)))
ds <- data.frame(var, group)
ggplot(data = ds, mapping = aes(x = var)) +
  geom_density(aes(color = group))
```



Summary Stats

```r
favstats(var ~ group, data = ds)
```

```
##   group      min       Q1   median       Q3       max     mean        sd  n
## 1     1 41.94485 46.97295 50.06203 53.72178  59.72196 50.57180  4.593161 25
## 2     2 31.46573 43.51127 46.25651 51.25870  58.76064 46.17537  6.220347 25
## 3     3 16.45020 46.52976 52.74248 56.64019 112.94027 54.68582 23.653932 25
```

```
##   missing
## 1       0
## 2       0
## 3       0
```

Now we look at the test procedures. Theoretically, testing X vs. Y, or X vs. Z, or Y vs. Z should ALL result in rejections of null hypotheses of equal centers, but in some cases we have a shift model and in other cases we don't.

Wilcoxon Rank Sum Procedures

```r
wilcox.test(X, Y) # shift model valid
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  X and Y
## W = 452, p-value = 0.006236
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(X, Z) # shift model not valid
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  X and Z
## W = 290, p-value = 0.6721
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(Y, Z) # shift model not valid
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  Y and Z
## W = 213, p-value = 0.05422
## alternative hypothesis: true location shift is not equal to 0
```

What do you notice about these test results? For which tests do you reject the null hypothesis at reasonable significance levels?

SOLUTION

When comparing X and Y, the test is able to reject the null hypothesis of no shift in the two distributions. It comes close to rejecting the null hypothesis of no shift (at alpha = 0.05) for Y vs Z, but cannot distinguish X and Z. In other words, it worked for the valid shift model and not otherwise.

T-test

```r
t.test(X, Y)
```

```
##
##  Welch Two Sample t-test
##
## data:  X and Y
## t = 2.8429, df = 44.174, p-value = 0.006745
## alternative hypothesis: true difference in means is not equal to 0
```

```
## 95 percent confidence interval:
##   1.280050 7.512799
## sample estimates:
## mean of x mean of y
##  50.57180  46.17537
```

```r
t.test(X, Z)
```

```
##
##  Welch Two Sample t-test
##
## data:  X and Z
## t = -0.85368, df = 25.807, p-value = 0.4011
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -14.023532   5.795487
## sample estimates:
## mean of x mean of y
##  50.57180  54.68582
```

```r
t.test(Y, Z)
```

```
##
##  Welch Two Sample t-test
##
## data:  Y and Z
## t = -1.7398, df = 27.304, p-value = 0.09316
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -18.54202   1.52113
## sample estimates:
## mean of x mean of y
##  46.17537  54.68582
```

What do you notice about these test results? For which tests do you reject the null hypothesis at reasonable significance levels?

SOLUTION

The t-test has the same issues as the Rank Sum test. Distinguishing the centers of the distributions for X and Y seems easy (and is the shift model), while distinguishing Y and Z is doable at higher alphas. X and Z seem to have similar centers, despite being generated in a way to have different centers.

van der Waerden's Test

```r
testXY <- waerden.test(c(X, Y), rep(1:2, each = 25))  # be sure agricolae is loaded
testXY$statistics # p-value is middle entry
```

```
##      Chisq Df     p.chisq  t.value       MSD
##   7.603675  1 0.005824944 2.010635 0.4974262
```

```r
testXZ <- waerden.test(c(X, Z), rep(1:2, each = 25))
testXZ$statistics # p-value is middle entry
```

```
##       Chisq Df   p.chisq  t.value       MSD
##   0.1200563  1 0.7289734 2.010635 0.5405216
```

```r
testYZ <- waerden.test(c(Y, Z), rep(1:2, each = 25))
testYZ$statistics # p-value is middle entry
```

```
##      Chisq Df    p.chisq  t.value       MSD
##   2.876776  1 0.08986578 2.010635 0.5250584
```

What do you notice about these test results? For which tests do you reject the null hypothesis at reasonable significance levels?

SOLUTION

This test has the same issues as the Rank Sum test and t-test. Distinguishing the centers of the distributions for X and Y seems easy (shift model is valid), while distinguishing Y and Z is doable at higher alphas. X and Z seem to have similar centers, despite being generated in a way to have different centers.

Overall, the shift model appropriateness doesn't seem to hamper these nonparametric tests. They reach the same conclusions as the t-test in each setting we investigated.

Nonparametric Methods in Literature

As you learn new nonparametric methods, you are likely curious about how they are used in practice. Specifically, how are they used in application areas of interest to you? Let's explore how the rank sum test is used.

Use Google Scholar and do a search for "rank sum test" (or "Wilcoxon rank sum" or "Mann-Whitney U") and an application area of interest to you. For example, search for ' "rank sum" archaeology' to look for applications of the rank sum test in archaeology. (If the subject area has multiple words, you may want to put quotes around it as well.) Choose an article that pops up from your search that is interesting to you (more recent than 2010 would be best, if possible) and answer the questions below.

If you are having issues picking an application area or choosing an article, some potential articles are provided below. Applications do not need to be in journals in statistics!

April Batson Malone, "The Effects of Live Music on the Distress of Pediatric Patients Receiving Intravenous Starts, Venipunctures, Injections, and Heel Sticks," Journal of Music Therapy, 33.1, Spring 1996: 19–33, https://doi.org/10.1093/jmt/33.1.19.

Guilbeault, Douglas, Joshua Becker, and Damon Centola, "Social learning and partisan bias in the interpretation of climate trends," Proceedings of the National Academy of Sciences, 115.39, 2018: 9714-9719.

Padmanathan, Prianka, et al., "Social media use, economic recession and income inequality in relation to trends in youth suicide in high-income countries: a time trends analysis," Journal of affective disorders, 275, 2020: 58-65.

Shinde, Vasant S., et al., "Archaeological and anthropological studies on the Harappan cemetery of Rakhigarhi, India," PLoS One, 13.2, 2018: e0192299.

Provide a citation for your article. Google Scholar should be able to provide a complete citation for you to copy/paste.

SOLUTION:

Answers will vary.

Summarize what the article is about in a few sentences.

SOLUTION:

Answers will vary.

What was the rank sum test used for in the article? Describe as best you can. For example, what were the hypotheses being tested?

SOLUTION:

Answers will vary.

Does the article state why the rank sum test was used? If so, what reason do they give?

SOLUTION:

Answers will vary.

Are any other statistical procedures mentioned in the paper, that you can tell? If so, what else is mentioned?

SOLUTION:

Answers will vary.

Data Sources

Day1Survey data is from the Stat2Data pkg in R.

Salaries data is from the carData pkg in R.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to Nonparametric Statistical Methods (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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